ABHISHEK TIWARI

abhiofficial1611@gmail.com | +91-9452371697 | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Full-Stack Developer specializing in building and maintaining web applications using React.js, Django, Python, JavaScript, and SQL databases. Currently developing a Management Information System (MIS), contributing to frontend development, backend API development, database design, and API integration. Proficient in designing REST APIs and working with MySQL, PostgreSQL, Git, Jira, and Postman within Agile development environments.

SKILLS

- **Programming Languages:** Python, JavaScript
- **Frameworks & Libraries:** React.js, Node.js, Express.js, Django, JWT Authentication
- **Tools:** Git, Jira, Postman, IntelliJ, Jupyter Notebook, VS Code
- **Databases:** MySQL, PostgreSQL
- **Methodologies:** Agile, Scrum, SDLC
- **Specializations:** Full-Stack Development, Backend Development, Frontend Development

PROJECTS

Netflix Recommendation System

Python · TensorFlow · REST API · React · Flask

- Cleaned and pre-processed a dataset of over 17,000 Netflix titles, addressing missing values across 10+ feature columns to ensure data quality.
- Developed a TF-IDF content similarity engine, reducing recommendation latency to 200ms and delivering personalized results across more than 5,000 titles.
- Implemented collaborative filtering techniques, achieving 85% precision in predicted user ratings during offline evaluations to enhance user experience.
- Delivered an intuitive React frontend featuring over 3 UI components, successfully integrating 2 machine learning models through REST API calls for optimal performance.

Fake News Detection

Python · TensorFlow · Scikit-learn

- Trained a classification model on 10,000+ news articles, achieving 95% accuracy on validation sets
- Implemented NLP preprocessing techniques such as tokenization, stopword removal, and TF-IDF
- Reduced feature dimensionality by 60% through effective data preprocessing
- Engineered 15+ insightful text-based features, including sentiment score, word count, and n-gram frequency

Fabric Classification System

Python · TensorFlow · Keras · MobileNet · OpenCV · Streamlit · Grad-CAM · Matplotlib

- Developed a lightweight CNN-based fabric classification system using a modified MobileNet architecture to differentiate between cotton and blended fabrics, achieving an impressive 85.71% test accuracy.
- Optimized the model for deployment, reducing architecture size to 1.09M parameters (~5MB) while preserving performance levels comparable to larger models such as ResNet50 and VGG16.
- Constructed a comprehensive preprocessing and augmentation pipeline utilizing Keras ImageDataGenerator, effectively expanding the training dataset by 63× to enhance model generalization capabilities.
- Created a Streamlit web application featuring real-time predictions, confidence visualization, and Grad-CAM explainability heatmaps, aiming to support interpretable AI-based classification.

EDUCATION

Bachelors of Technology, Computer Engineering

Pranveer Singh Insitute of Technology

December **2022 - August 2026 | Kanpur, India**

CGPA: 7.1

CERTIFICATIONS

- CS50 - Harvard University (edX)
- Machine Learning Specialization - Stanford University
- SQL - DataCamp